MANELISI LUTHULI

Aspiring Software Engineer
BSc Computer Science & Mathematics — University of Cape Town
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Technical Skills

Languages	Java, Python, JavaScript
Tools & Technologies	Git, Unix/Linux, VS Code, Maven, Spring Boot, MongoDB
Other	HTML/CSS, REST APIs, Socket Programming, Multi-threading, Concurrency, Binary File Handling

Projects

Peer-to-Peer File Sharing Application *Java — Sockets & Networking*

- Built a complete P2P system using Java TCP/IP sockets with multi-threading for independent file transfers.
- Designed a custom communication protocol supporting real-time broadcasting to all connected peers.
- Implemented safe shared-resource management across threads using synchronisation techniques.

Concurrent Grid Processing System *Java*

- Developed a grid-based system to explore parallelism and concurrency using Java threads.
- Compared sequential vs. parallel performance and analysed scalability relative to CPU cores.
- Resolved race conditions through proper synchronisation mechanisms.

Sorting Algorithm Performance Analysis *Java / Statistics*

- Designed experiments comparing multiple sorting algorithms across varying dataset sizes and distributions.
- Analysed runtime performance using statistical methods and visualised results with graphs and reports.
- Integrated the Random.org API for truly random sequence generation with reliable response handling.

WAVE File Processing *MIPS Assembly*

- Developed a low-level program to read, manipulate, and reverse audio data in WAVE files.
- Worked directly with binary file structures and memory handling.

Excel Data Automation Tool *Python — Apache POI*

- Created a tool to automate writing and structuring data into Excel files.
- Significantly improved efficiency for repetitive data-handling tasks.

Binary Search Tree Application *Java — Data Structures*

- Built a proof-of-concept social media-like system (inspired by TikTok) using BSTs as the core data structure.

- Implemented multiple on-demand operations with in-memory data storage.

Dijkstra's Shortest Paths Algorithm *Java — Algorithms*

- Programmatically implemented Dijkstra's algorithm and compared performance against theoretical bounds.
- Included custom dataset generation for comprehensive testing.

Education

University of Cape Town *2025*
BSc in Computer Science & Mathematics

Relevant Coursework: Databases, Data Structures & Algorithms, Software Engineering, Networking, Concurrency & Parallelism, Computer Architecture, Operating Systems, Graph Theory, Linear Algebra, Real Analysis, Discrete Mathematics.

Holy Family College, Durban
Dux Scholar (Top Academic Achiever)